

Counterfactual Statistical Framework

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a Counterfactual Statistical Framework in which the primitive inferential object is not merely the observed dataset, a sampling distribution, a likelihood, or a posterior, but a structured family of admissible datasets that could have been observed under plausible unrealized worlds. The framework formalizes counterfactual datasets, admissibility relations, counterfactual generators, inference orbits, counterfactual estimands, stability indices, fragility measures, counterfactual confidence sets, counterfactual likelihood, Bayesian counterfactual posteriors, and counterfactual decision risk. The framework is designed for domains in which observed evidence is historically contingent, strategically endogenous, and structurally path-dependent: economics, finance, political science, geopolitics, international relations, warfare economics, military science, crisis analysis, statistics, data science, and artificial intelligence. The central claim is that an inference is credible not only when it is significant, unbiased, efficient, predictive, or decision-theoretically optimal in the realized world, but when it survives admissible counterfactual worlds.

The paper ends with “The End”

Keywords: counterfactual statistics, causal inference, robust statistics, path dependence, structural uncertainty, statistical fragility, decision theory, political economy, finance, warfare economics, artificial intelligence.

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1 Introduction

Classical statistical inference studies how observed data support statements about unknown parameters, distributions, hypotheses, predictions, or decisions. In its most familiar form, one observes a dataset

$$D = \{Z_i\}_{i=1}^n,$$

specifies a statistical functional

$$T : \mathcal{D} \rightarrow \Theta,$$

and reports

$$T(D).$$

Uncertainty is then attached through sampling distributions, likelihoods, confidence sets, posteriors, asymptotic approximations, bootstrap distributions, or predictive loss.

The Counterfactual Statistical Framework begins from a different premise. The observed dataset is only one realized branch of a broader space of plausible unrealized empirical histories. The central object is therefore not merely

$$T(D),$$

but the counterfactual orbit

$$\mathcal{O}_T(D) = \{T(D') : D' \in \mathcal{C}(D)\},$$

where $\mathcal{C}(D)$ is the set of admissible counterfactual datasets associated with the observed dataset D .

The motivating question is:

Would the same statistical conclusion have emerged under plausible alternative empirical worlds?

This question is different from ordinary sampling uncertainty. It is also different from causal treatment counterfactuals alone. It concerns the stability of the entire inferential conclusion across admissible unrealized data histories.

The framework is particularly relevant in settings where the observed world is unique, non-repeatable, and path-dependent:

macroeconomic crises, wars, elections, financial crashes, coups, sanctions, pandemics, strategic deterrence, tec

In such settings, the realized data are not merely noisy. They are historically contingent.

2 Relation to Existing Statistical Traditions

Counterfactual statistics draws on, but is distinct from, several established traditions.

Table 1: Counterfactual Statistics and Neighboring Fields

Field	Primary object	Counterfactual statistical extension
Classical inference	Sampling distribution of $T(D)$	Inference orbit over admissible unrealized datasets
Bayesian inference	Posterior $\pi(\theta \mid D)$	Posterior induced by a counterfactual likelihood over $\mathcal{C}(D)$
Bootstrap	Resampled datasets from D	Structurally admissible alternative worlds, not merely resamples
Robust statistics	Resistance to contamination	Stability under historically or institutionally plausible perturbations
Causal inference	Potential outcomes under interventions	Stability of full statistical conclusions across possible empirical histories
Sensitivity analysis	Variation in assumptions	Variation in admissible worlds, mechanisms, and data-generating histories
Distributionally robust optimization	Worst-case distributions	Worst-case or average-case loss over admissible counterfactual datasets
Scenario analysis and wargaming	Alternative scenarios	Formal statistical inference over scenario-generated datasets

The distinctive contribution of the framework is the systematic treatment of counterfactual datasets as inferential objects.

3 Basic Objects

Let $\mathcal{D}$ denote the space of possible datasets. An observed dataset is

$$D \in \mathcal{D}.$$

Let $\mathcal{W}$ denote a space of possible worlds. Each world $w \in \mathcal{W}$ induces a dataset

$$D_w \in \mathcal{D}.$$

The observed world is denoted w_0 , and

$$D = D_{w_0}.$$

Definition 1 (Counterfactual Statistical Problem). *A counterfactual statistical problem is a tuple*

$$\mathfrak{P} = (D, \mathcal{D}, \mathcal{W}, \mathcal{C}, \mathcal{Q}, T, S, L),$$

where:

- $D \in \mathcal{D}$ is the observed dataset;
- $\mathcal{W}$ is the space of possible worlds;
- $\mathcal{C}(D) \subseteq \mathcal{D}$ is the set of admissible counterfactual datasets;
- $\mathcal{Q}_D$ is a probability law over $\mathcal{C}(D)$;
- $T : \mathcal{D} \rightarrow \Theta$ is a statistical functional;
- $S : \mathcal{D} \rightarrow \mathcal{A}$ is a substantive conclusion map;
- $L : \mathcal{A} \times \mathcal{D} \rightarrow \mathbb{R}$ is a loss function.

The observed statistic is

$$T_0 = T(D).$$

The counterfactual orbit of T at D is

$$\mathcal{O}_T(D) = \{T(D') : D' \in \mathcal{C}(D)\}.$$

Definition 2 (Counterfactual Orbit). *The counterfactual orbit of an inferential functional T is the set*

$$\mathcal{O}_T(D) = \{T(D') : D' \in \mathcal{C}(D)\}.$$

It is the collection of conclusions that the same inferential procedure would have generated across admissible counterfactual datasets.

4 Admissibility

The central constraint is that counterfactual datasets must not be arbitrary. They must be admissible.

Definition 3 (Counterfactual Admissibility Relation). *Let $\sim_{\text{cf}}$ be a binary relation on $\mathcal{D}$. We write*

$$D' \sim_{\text{cf}} D$$

when D' is a counterfactually admissible alternative to D . Then

$$\mathcal{C}(D) = \{D' \in \mathcal{D} : D' \sim_{\text{cf}} D\}.$$

A valid counterfactual dataset must preserve the domain structure of the empirical problem. For a panel dataset, a counterfactual object should remain a panel. For a strategic crisis, the counterfactual object should preserve the strategic actors unless actor replacement is explicitly part of the counterfactual operation. For a financial time series, the counterfactual object should respect market calendars, liquidity constraints, and feasible price paths.

Axiom 1 (Observed-World Embedding). *The observed dataset is admissible relative to itself:*

$$D \in \mathcal{C}(D).$$

Axiom 2 (Admissibility Constraint). *There exists a predicate*

$$\Phi : \mathcal{D} \times \mathcal{D} \rightarrow \{0, 1\}$$

such that

$$D' \in \mathcal{C}(D) \implies \Phi(D', D) = 1.$$

Axiom 3 (Generator Transparency). *The mechanism used to construct $\mathcal{C}(D)$ must be explicitly declared.*

The admissibility predicate Φ may encode statistical, economic, institutional, geopolitical, or physical restrictions:

$$\Phi(D', D) = \Phi_{\text{stat}}(D', D)\Phi_{\text{econ}}(D', D)\Phi_{\text{inst}}(D', D)\Phi_{\text{geo}}(D', D)\Phi_{\text{war}}(D', D).$$

5 Counterfactual Generators

Definition 4 (Counterfactual Generator). *A deterministic counterfactual generator is a mapping*

$$G : \mathcal{D} \rightarrow 2^{\mathcal{D}},$$

with

$$G(D) = \mathcal{C}(D).$$

A stochastic counterfactual generator is a Markov kernel

$$\mathcal{Q}_D(A) = \Pr(D' \in A \mid D), \quad A \subseteq \mathcal{D}.$$

A counterfactual generator may operate at different levels.

5.1 Observation-Level Counterfactuals

At the observation level,

$$Z'_i = h_i(Z_i, \varepsilon_i),$$

where ε_i encodes measurement error, missingness, misclassification, or local perturbations.

5.2 Sample-Level Counterfactuals

At the sample level,

$$D' = \{Z'_i : i \in I'\},$$

where I' represents an admissible alternative sample, attrition pattern, treatment allocation, or selection mechanism.

5.3 Process-Level Counterfactuals

At the process level,

$$P' \in \mathcal{P}_{\text{cf}}(P),$$

where $\mathcal{P}_{\text{cf}}(P)$ is a neighborhood of admissible data-generating processes.

5.4 World-Level Counterfactuals

At the world level,

$$w' \in \mathcal{W}_D \subseteq \mathcal{W}, \quad D' = D_{w'}.$$

These counterfactuals may encode alternative policy timing, alliance behavior, military logistics, financial liquidity, technological diffusion, weather shocks, or institutional breakdown.

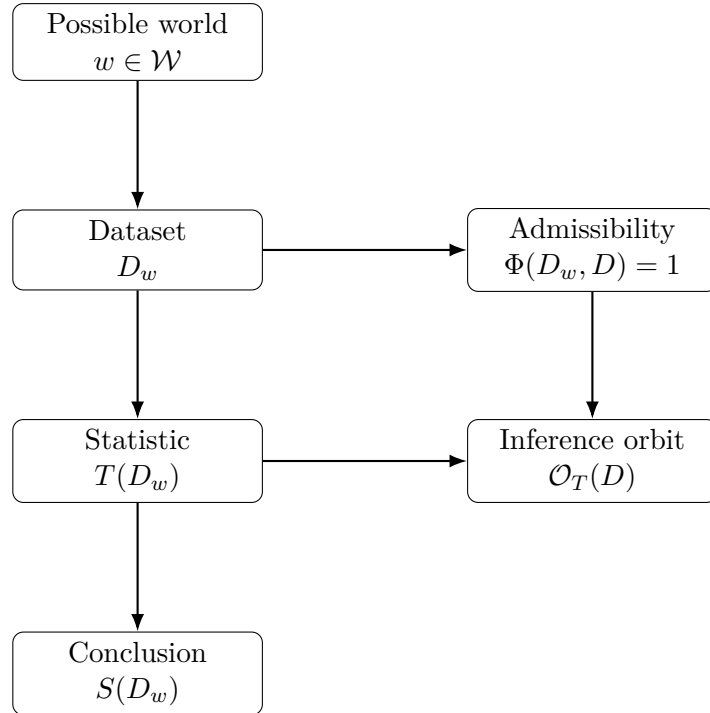


Figure 1: Counterfactual Statistical Pipeline

6 Counterfactual Estimands

Let $\theta(P)$ denote a conventional estimand for a data-generating process P . In a possible-world formulation, each world w has an associated process P_w , yielding

$$\theta_w = \theta(P_w).$$

The counterfactual estimand set is

$$\Theta_{\text{cf}}(D) = \{\theta(P_w) : w \in \mathcal{W}_D\}.$$

Given a statistic $T(D')$, define the counterfactual mean:

$$\bar{T}_{\text{cf}}(D) = \mathbb{E}_{D' \sim \mathcal{Q}_D}[T(D')].$$

The counterfactual variance is

$$V_{\text{cf}}(T; D) = \text{Var}_{D' \sim \mathcal{Q}_D}(T(D')).$$

The counterfactual range is

$$R_{\text{cf}}(T; D) = \sup_{D' \in \mathcal{C}(D)} T(D') - \inf_{D' \in \mathcal{C}(D)} T(D').$$

For scalar T , define normalized inferential fragility:

$$F_{\text{cf}}(T; D) = \frac{R_{\text{cf}}(T; D)}{|T(D)| + \varepsilon}, \quad \varepsilon > 0.$$

7 Counterfactual Stability

Let $S : \mathcal{D} \rightarrow \mathcal{A}$ be a conclusion map. Examples include:

$$S(D) = \begin{cases} 1, & \text{if the treatment effect is positive and statistically significant,} \\ 0, & \text{otherwise,} \end{cases}$$

or

$$S(D) = \begin{cases} \text{attack,} \\ \text{deter,} \\ \text{negotiate.} \end{cases}$$

Definition 5 (Counterfactual Stability). *The counterfactual stability of conclusion $S(D)$ is*

$$\text{Stab}_{\text{cf}}(S; D) = \Pr_{D' \sim \mathcal{Q}_D} \{S(D') = S(D)\}.$$

If $\mathcal{C}(D)$ is finite and equally weighted, then

$$\text{Stab}_{\text{cf}}(S; D) = \frac{1}{|\mathcal{C}(D)|} \sum_{D' \in \mathcal{C}(D)} \mathbf{1}\{S(D') = S(D)\}.$$

Definition 6 (Counterfactual Fragility). *Counterfactual fragility is*

$$\text{Frag}_{\text{cf}}(S; D) = 1 - \text{Stab}_{\text{cf}}(S; D).$$

Thus a result may satisfy ordinary statistical significance while still being counterfactually fragile:

$$p_{\text{classical}} = 0.02, \quad \text{Frag}_{\text{cf}} = 0.47.$$

8 Counterfactual Confidence Sets

Let $CI(D)$ be a conventional confidence set computed from D .

Definition 7 (Weak Counterfactual Confidence Set). *The weak counterfactual confidence set is*

$$CI_{\text{cf}}^{\cup}(D) = \bigcup_{D' \in \mathcal{C}(D)} CI(D').$$

It contains parameter values that are plausible in at least one admissible counterfactual world.

Definition 8 (Strong Counterfactual Confidence Set). *The strong counterfactual confidence set is*

$$CI_{\text{cf}}^{\cap}(D) = \bigcap_{D' \in \mathcal{C}(D)} CI(D').$$

It contains parameter values that remain plausible across all admissible counterfactual worlds.

The conceptual distinction is:

$$CI_{\text{cf}}^{\cup}(D) = \text{possible under some admissible worlds,}$$

whereas

$$CI_{\text{cf}}^{\cap}(D) = \text{robust under all admissible worlds.}$$

9 Counterfactual Likelihood and Posterior

Let $p_\theta(D)$ be an ordinary likelihood.

Definition 9 (Counterfactual Log-Likelihood). *The counterfactual log-likelihood is*

$$\ell_{\text{cf}}(\theta; D) = \mathbb{E}_{D' \sim \mathcal{Q}_D} [\log p_\theta(D')].$$

If $\mathcal{C}(D)$ is finite,

$$\ell_{\text{cf}}(\theta; D) = \sum_{D' \in \mathcal{C}(D)} \omega(D') \log p_\theta(D'), \quad \sum_{D'} \omega(D') = 1.$$

The counterfactual maximum likelihood estimator is

$$\hat{\theta}_{\text{cf}} = \arg \max_{\theta \in \Theta} \ell_{\text{cf}}(\theta; D).$$

Given a prior $\pi(\theta)$, the counterfactual posterior is

$$\pi_{\text{cf}}(\theta \mid D) \propto \exp\{\ell_{\text{cf}}(\theta; D)\} \pi(\theta).$$

This posterior favors parameters that explain the admissible counterfactual neighborhood of D , not merely the realized dataset.

10 Counterfactual Risk and Decision Theory

Let $a \in \mathcal{A}$ be an action and $L(a, D')$ a loss function.

Definition 10 (Counterfactual Risk). *The counterfactual risk of action a is*

$$R_{\text{cf}}(a; D) = \mathbb{E}_{D' \sim \mathcal{Q}_D} [L(a, D')].$$

The average-case counterfactual decision is

$$a_{\text{cf}}^* = \arg \min_{a \in \mathcal{A}} R_{\text{cf}}(a; D).$$

The minimax counterfactual decision is

$$a_{\text{mm}}^* = \arg \min_{a \in \mathcal{A}} \sup_{D' \in \mathcal{C}(D)} L(a, D').$$

This gives two decision criteria:

average robustness and worst-case robustness.

11 Structural Decomposition

A central objective is to distinguish structural signal from world-path accident. Suppose

$$T(D') = \mu_{\text{struct}} + \eta_{\text{world}}(D') + \varepsilon(D'),$$

where:

μ_{struct} = stable structural component,

$\eta_{\text{world}}(D')$ = counterfactual world-path component,

and

$$\varepsilon(D') = \text{ordinary statistical noise.}$$

Then

$$\text{Var}(T) = \text{Var}(\eta_{\text{world}}) + \text{Var}(\varepsilon) + 2 \text{Cov}(\eta_{\text{world}}, \varepsilon).$$

Define the structural share of inference:

$$\text{SSI}(T; D) = 1 - \frac{V_{\text{cf}}(T; D)}{V_{\text{total}}(T; D) + \varepsilon}, \quad \varepsilon > 0.$$

High SSI indicates that the conclusion is structurally stable. Low SSI indicates high path dependence.

12 Theorems

Theorem 1 (Threshold Stability Bound). *Let $T(D) \in \mathbb{R}$ and let*

$$S(D) = \mathbf{1}\{T(D) > c\}.$$

Assume that for every $D' \in \mathcal{C}(D)$,

$$|T(D') - T(D)| \leq \delta.$$

If

$$|T(D) - c| > \delta,$$

then

$$S(D') = S(D)$$

for every $D' \in \mathcal{C}(D)$, and therefore

$$\text{Stab}_{\text{cf}}(S; D) = 1.$$

Proof. Since

$$|T(D') - T(D)| \leq \delta,$$

we have

$$T(D') \in [T(D) - \delta, T(D) + \delta].$$

If $T(D) > c$ and $T(D) - c > \delta$, then

$$T(D') \geq T(D) - \delta > c.$$

Hence $S(D') = 1 = S(D)$. If $T(D) < c$ and $c - T(D) > \delta$, then

$$T(D') \leq T(D) + \delta < c.$$

Hence $S(D') = 0 = S(D)$. Thus the conclusion is invariant over $\mathcal{C}(D)$, proving the claim. $\square$

Theorem 2 (Lipschitz Counterfactual Stability). *Let $T : \mathcal{D} \rightarrow \mathbb{R}$ be Lipschitz with respect to a metric d :*

$$|T(D') - T(D)| \leq Kd(D', D).$$

Suppose

$$\sup_{D' \in \mathcal{C}(D)} d(D', D) \leq r.$$

If

$$|T(D) - c| > Kr,$$

then

$$\text{Stab}_{\text{cf}}(S; D) = 1$$

for $S(D) = \mathbf{1}\{T(D) > c\}$.

Proof. For all $D' \in \mathcal{C}(D)$,

$$|T(D') - T(D)| \leq Kd(D', D) \leq Kr.$$

Apply the threshold stability bound with $\delta = Kr$. □

Proposition 1 (Counterfactual Mean-Squared Decomposition). *Define*

$$\text{Bias}_{\text{cf}}(T; D) = \mathbb{E}_{\mathcal{Q}_D}[T(D')] - T(D),$$

and

$$\text{MSD}_{\text{cf}}(T; D) = \mathbb{E}_{\mathcal{Q}_D}[(T(D') - T(D))^2].$$

Then

$$\text{MSD}_{\text{cf}}(T; D) = V_{\text{cf}}(T; D) + \text{Bias}_{\text{cf}}(T; D)^2.$$

Proof. Let

$$\mu_{\text{cf}} = \mathbb{E}_{\mathcal{Q}_D}[T(D')].$$

Then

$$T(D') - T(D) = (T(D') - \mu_{\text{cf}}) + (\mu_{\text{cf}} - T(D)).$$

Squaring and taking expectations gives

$$\mathbb{E}[(T(D') - T(D))^2] = \mathbb{E}[(T(D') - \mu_{\text{cf}})^2] + (\mu_{\text{cf}} - T(D))^2,$$

because the cross term has expectation zero. This is the desired decomposition. □

The following space was deliberately left blank.

13 Algorithms

Algorithm 1: Counterfactual Statistical Inference

Input: observed dataset D , statistic T , conclusion map S , generator G , number of worlds M .

1. Generate counterfactual datasets:

$$D^{(1)}, \dots, D^{(M)} \sim G(D).$$

2. Compute observed statistic and conclusion:

$$T_0 = T(D), \quad S_0 = S(D).$$

3. For each $m = 1, \dots, M$, compute

$$T_m = T(D^{(m)}), \quad S_m = S(D^{(m)}).$$

4. Estimate counterfactual stability:

$$\widehat{\text{Stab}}_{\text{cf}} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{S_m = S_0\}.$$

5. Estimate fragility:

$$\widehat{\text{Frag}}_{\text{cf}} = 1 - \widehat{\text{Stab}}_{\text{cf}}.$$

6. Estimate counterfactual range:

$$\widehat{R}_{\text{cf}} = \max_m T_m - \min_m T_m.$$

7. Report

$$T_0, \quad \widehat{R}_{\text{cf}}, \quad \widehat{\text{Stab}}_{\text{cf}}, \quad \widehat{\text{Frag}}_{\text{cf}}.$$

Algorithm 2: Counterfactual Model Selection

Input: candidate models $\mathcal{M}_1, \dots, \mathcal{M}_K$, data D , generator G , loss L , number of worlds M .

1. Generate

$$D^{(1)}, \dots, D^{(M)} \sim G(D).$$

2. For each model $\mathcal{M}_k$ and world m , fit $\mathcal{M}_k$ on $D^{(m)}$.

3. Compute loss

$$L_{km} = L(\mathcal{M}_k, D^{(m)}).$$

4. Compute counterfactual risk

$$\widehat{R}_{\text{cf}}(\mathcal{M}_k) = \frac{1}{M} \sum_{m=1}^M L_{km}.$$

5. Select

$$\mathcal{M}_{\text{cf}}^* = \arg \min_{1 \leq k \leq K} \widehat{R}_{\text{cf}}(\mathcal{M}_k).$$

Algorithm 3: Counterfactual Stress Testing for AI Models

Input: training dataset D , learning algorithm A , test point x , generator G , number of worlds M .

1. Train observed model

$$f_0 = A(D).$$

2. Generate counterfactual training datasets

$$D^{(1)}, \dots, D^{(M)} \sim G(D).$$

3. Train counterfactual models

$$f_m = A(D^{(m)}).$$

4. Compute prediction instability:

$$I(x) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{f_m(x) \neq f_0(x)\}.$$

5. Declare prediction robust if

$$I(x) \leq \tau,$$

where τ is a tolerance threshold.

14 Economics

In economics, empirical conclusions often depend on historical paths, institutions, policy timing, market structure, and macroeconomic shocks.

Let Y_{it} be output, P_{it} a policy variable, and X_{it} controls. A conventional panel model is

$$Y_{it} = \alpha_i + \lambda_t + \beta P_{it} + X_{it}'\gamma + u_{it}.$$

The observed estimate is

$$\hat{\beta} = T(D).$$

A counterfactual economic generator may alter:

policy timing, commodity prices, exchange rates, sanction intensity, institutional constraints, monetary conditions.

The counterfactual policy-effect orbit is

$$\mathcal{O}_\beta(D) = \{\hat{\beta}(D') : D' \in \mathcal{C}(D)\}.$$

The policy conclusion

$$S(D) = \mathbf{1}\{\hat{\beta}(D) > 0\}$$

has stability

$$\text{Stab}_{\text{cf}}(S; D) = \Pr\{\hat{\beta}(D') > 0 \mid D' \sim \mathcal{Q}_D\}.$$

A policy should be considered structurally supported only when this probability is high.

15 Finance

In finance, a trading strategy may be profitable in one realized price path but fragile across alternative liquidity, volatility, and order-flow paths.

Let π denote a strategy and $R_t(\pi, D)$ its return at time t . The observed Sharpe ratio is

$$SR(\pi, D) = \frac{\mathbb{E}_t[R_t(\pi, D)]}{\sqrt{\text{Var}_t(R_t(\pi, D))}}.$$

The counterfactual Sharpe orbit is

$$\mathcal{O}_{SR}(\pi, D) = \{SR(\pi, D') : D' \in \mathcal{C}(D)\}.$$

Define counterfactual profitability stability:

$$\text{Stab}_{\text{profit}}(\pi; D) = \Pr_{D' \sim \mathcal{Q}_D} \{SR(\pi, D') > 0\}.$$

Define counterfactual drawdown risk:

$$DD_{\text{cf}}(\pi; D) = \Pr_{D' \sim \mathcal{Q}_D} \left\{ \max_t DD_t(\pi, D') > d \right\}.$$

A trading strategy is robust only when:

$$\text{Stab}_{\text{profit}}(\pi; D) \approx 1 \quad \text{and} \quad DD_{\text{cf}}(\pi; D) \approx 0.$$

16 Political Science, Geopolitics, and International Relations

Political datasets are often produced by non-repeatable sequences of elections, elite bargains, protests, sanctions, coups, wars, and diplomatic signals.

Let Y_{ct} be regime stability for country c at time t , and let S_{ct} denote sanctions. A simple model is

$$Y_{ct} = \alpha_c + \lambda_t + \beta S_{ct} + X'_{ct}\gamma + \varepsilon_{ct}.$$

A counterfactual geopolitical generator may vary:

commodity prices, elite defections, alliance support, protest coordination, military loyalty, and foreign intervention.

The conclusion

$$S(D) = \mathbf{1}\{\hat{\beta} < 0\}$$

states that sanctions reduce regime stability. Its counterfactual support is

$$\Pr_{\text{cf}}(\hat{\beta} < 0) = \Pr_{D' \sim \mathcal{Q}_D} \{\hat{\beta}(D') < 0\}.$$

This allows one to distinguish:

sanctions worked in the realized world

from

sanctions are structurally destabilizing across plausible worlds.

17 Warfare Economics and Military Science

War is a domain of radical contingency. Outcomes depend on force ratios, logistics, morale, weather, deception, command quality, intelligence accuracy, industrial capacity, and alliance credibility.

Let $V_A(D)$ be an indicator that side A wins in dataset or simulated world D . The counterfactual victory probability is

$$P_{\text{cf}}(A \text{ wins}) = \Pr_{D' \sim \mathcal{Q}_D} \{V_A(D') = 1\}.$$

A victory in the observed world is structurally robust if

$$P_{\text{cf}}(A \text{ wins}) \approx 1.$$

It is contingent if

$$P_{\text{cf}}(A \text{ wins}) \approx \frac{1}{2}.$$

Warfare economics introduces resource constraints. Let B_A and B_B be war budgets, K_A and K_B capital stocks, M_A and M_B mobilized manpower, and Λ_A, Λ_B logistics capacities. A stylized combat production function is

$$C_j = A_j B_j^\alpha K_j^\kappa M_j^\mu \Lambda_j^\lambda, \quad j \in \{A, B\}.$$

The victory condition may be modeled as

$$V_A(D) = \mathbf{1} \left\{ \frac{C_A}{C_B} > \rho \right\},$$

where ρ is a theater-specific threshold.

Counterfactual warfare statistics studies

$$\Pr_{\text{cf}} \left(\frac{C_A(D')}{C_B(D')} > \rho \right).$$

18 AI, Machine Learning, and Data Science

In machine learning, the training dataset is itself historically contingent. A model trained on one realized dataset may behave differently under plausible alternative training histories.

Let A be a learning algorithm and

$$f_D = A(D)$$

the trained model. For a prediction point x , define prediction instability:

$$I(x) = \Pr_{D' \sim \mathcal{Q}_D} \{f_{D'}(x) \neq f_D(x)\}.$$

For regression, define prediction dispersion:

$$V_{\text{pred}}(x) = \text{Var}_{D' \sim \mathcal{Q}_D}(f_{D'}(x)).$$

For classification, define counterfactual class probability:

$$p_k^{\text{cf}}(x) = \Pr_{D' \sim \mathcal{Q}_D} \{f_{D'}(x) = k\}.$$

A classification is stable if

$$\max_k p_k^{\text{cf}}(x)$$

is close to one.

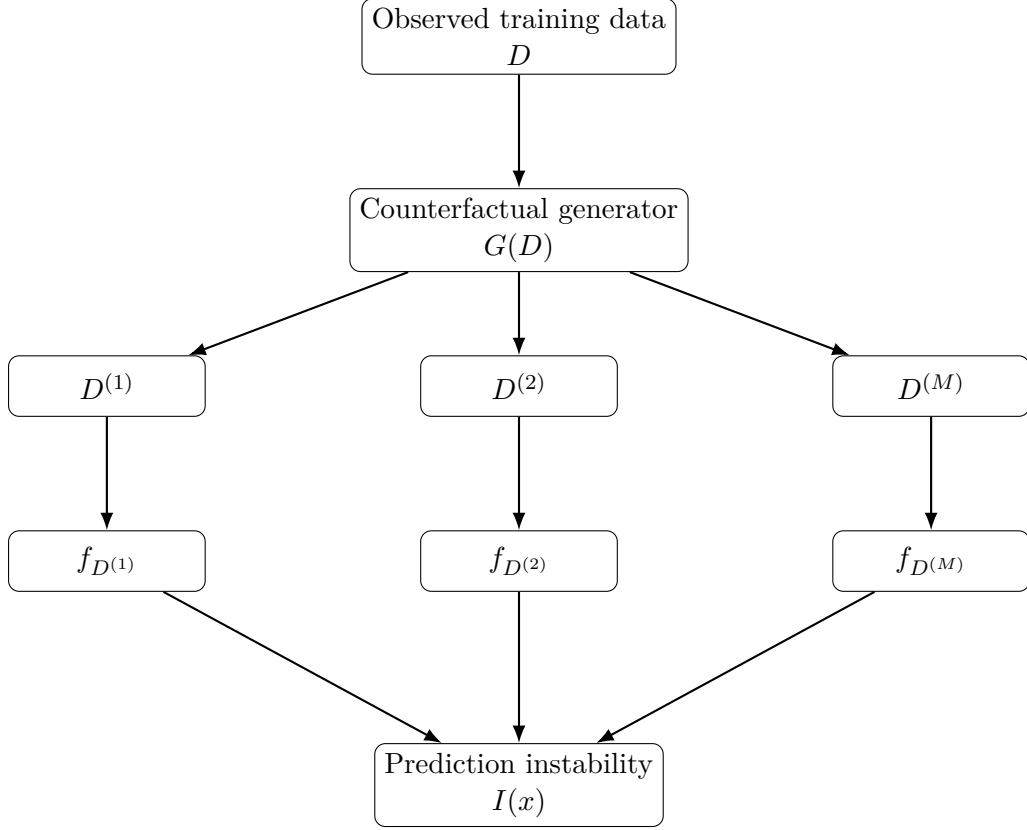


Figure 2: Counterfactual Stress Testing for Machine Learning

19 Reporting Standards

A counterfactual statistical report should not stop at:

$$\hat{\theta} = 0.41, \quad p = 0.03.$$

It should report:

$$\begin{aligned} \hat{\theta}_{\text{obs}} &= 0.41, \\ CI_{\text{obs}} &= [0.05, 0.77], \\ \Theta_{\text{cf}} &= [-0.12, 0.84], \\ \text{Stab}_{\text{cf}} &= 0.68, \\ \text{Frag}_{\text{cf}} &= 0.32, \\ p_{\text{cf}} &= 0.32. \end{aligned}$$

20 A Unified Interpretation

The framework can be summarized as:

Inference is credible to the extent that it survives admissible counterfactual worlds.

Equivalently:

The object of inference is not $T(D)$ alone, but $\{T(D') : D' \in \mathcal{C}(D)\}$.

This shifts statistical attention from realized-world fit to counterfactual stability.

Table 2: Suggested Counterfactual Statistical Reporting Template

Quantity	Notation	Interpretation
Observed estimate	$T(D)$	Realized-world estimate
Classical uncertainty	$CI(D)$	Sampling uncertainty
Counterfactual orbit	$\mathcal{O}_T(D)$	Inference across admissible worlds
Counterfactual range	$R_{\text{cf}}(T; D)$	Maximum inferential displacement
Stability	Stab_{cf}	Probability conclusion survives
Fragility	Frag_{cf}	Probability conclusion changes
Weak confidence set	$CI_{\text{cf}}^{\cup}$	Plausible in some worlds
Strong confidence set	$CI_{\text{cf}}^{\cap}$	Plausible in all worlds
Decision risk	$R_{\text{cf}}(a; D)$	Expected counterfactual loss

21 Conclusion

The Counterfactual Statistical Framework formalizes a mode of reasoning that is implicit in serious empirical work but rarely treated as a first-class statistical paradigm. In economics, the question is whether a policy effect survives alternative macroeconomic histories. In finance, it is whether alpha survives alternative price paths. In political science, it is whether a regime conclusion survives alternative elite, protest, sanction, and alliance dynamics. In warfare economics, it is whether victory reflects structural superiority or historical contingency. In artificial intelligence, it is whether a prediction survives plausible alternative training histories.

The central distinction is between:

a conclusion true in the observed branch

and

a conclusion stable across admissible branches.

Counterfactual statistics therefore complements classical inference, Bayesian inference, robust statistics, causal inference, simulation, machine learning validation, and decision theory. Its distinctive object is the admissible family of unrealized datasets, and its distinctive virtue is counterfactual stability.

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Glossary

Admissible counterfactual dataset

A dataset that could plausibly have been observed under a specified alternative world while preserving the structural constraints of the problem.

Admissibility predicate

A function $\Phi(D', D)$ that determines whether D' is a valid counterfactual alternative to D .

Counterfactual generator

A deterministic or stochastic rule that produces admissible counterfactual datasets from an observed dataset.

Counterfactual orbit

The set $\mathcal{O}_T(D) = \{T(D') : D' \in \mathcal{C}(D)\}$ of all inferential outputs across admissible counterfactual datasets.

Counterfactual stability

The probability that a substantive conclusion remains unchanged across admissible counterfactual datasets.

Counterfactual fragility

The probability that a substantive conclusion changes across admissible counterfactual datasets.

Counterfactual confidence set

A confidence object formed by unioning or intersecting conventional confidence sets over counterfactual datasets.

Counterfactual likelihood

A likelihood-like criterion obtained by averaging log-likelihood over admissible counterfactual datasets.

Counterfactual risk

The expected loss of an action over admissible counterfactual datasets.

Inference orbit

Another term for the counterfactual orbit of a statistic, estimator, test, classifier, or decision rule.

Path dependence

The dependence of an empirical conclusion on the realized historical trajectory rather than on stable structure.

Structural share of inference

A measure of how much of an inferential conclusion appears to be stable structure rather than counterfactual world-path variation.

World-path uncertainty

Uncertainty arising from the fact that the observed world is one historically contingent realization among many plausible alternatives.

The End